Most secretory and membrane-bound proteins participate in the *N*-linked glycosylation pathway, in which a complex polysaccharide group (also called a glycan) is linked to one or more asparagine residues on proteins in the lumen of the endoplasmic reticulum (ER). Over 30 congenital disorders are associated with defects in the enzymes responsible for the synthesis of *N*-linked glycans, and each disorder has a broad range of symptoms including eye, neuron, and bone pathologies.

The glycan is assembled on a dolichol phosphate molecule embedded in the ER membrane and consists primarily of *N*-acetylglucosamine (GlcNAc) and mannose residues. The anomeric carbon of each monosaccharide forms a glycosidic bond with the growing chain, and the full glycan is then transferred to a protein as a single unit. The first four steps of glycan synthesis are depicted in Figure 1.

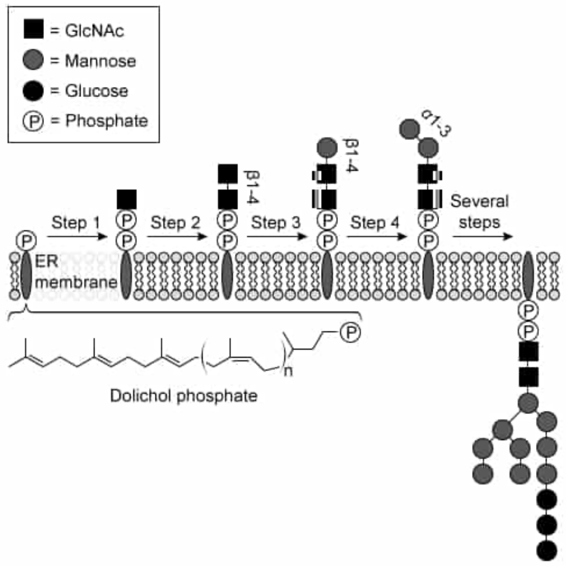


Figure 1 The first four steps of glycan synthesis in the *N*-linked glycosylation pathway and the full glycan. The structure of dolichol phosphate and linkages between carbohydrates are also shown.

Before a monosaccharide can be incorporated into the glycan, it must be activated by reacting with a nucleotide triphosphate (NTP) to form an NDP-carbohydrate molecule. The general mechanisms for formation of NDP-GlcNAc and NDP-mannose are shown in Figure 2, using the six-carbon sugar fructose 6-phosphate as the starting material.

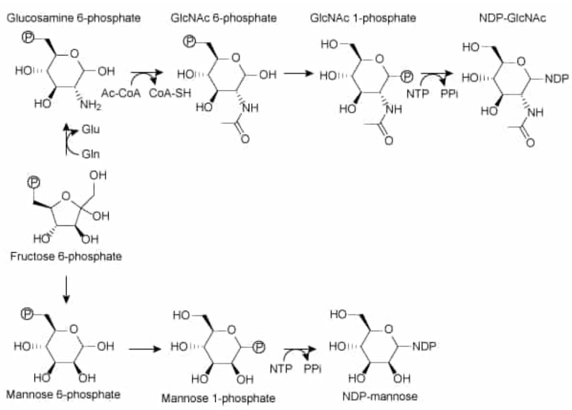


Figure 2 Formation of NDP-GlcNAc and NDP-mannose from fructose 6-phosphate

To determine which NDP activates each monosaccharide, researchers tested each step individually by incubating liposomes linked to precursor glycans, the appropriate transferase enzyme, and different NDP-carbohydrate molecules. Each test was done in the presence and absence of the drug tunicamycin. In each case, the anomeric carbon of the activated monosaccharide was a ^{14}C atom. After incubation, the liposomes were purified and analyzed by TLC and autoradiography. The results for the first four steps of the pathway are shown in Figure 3.

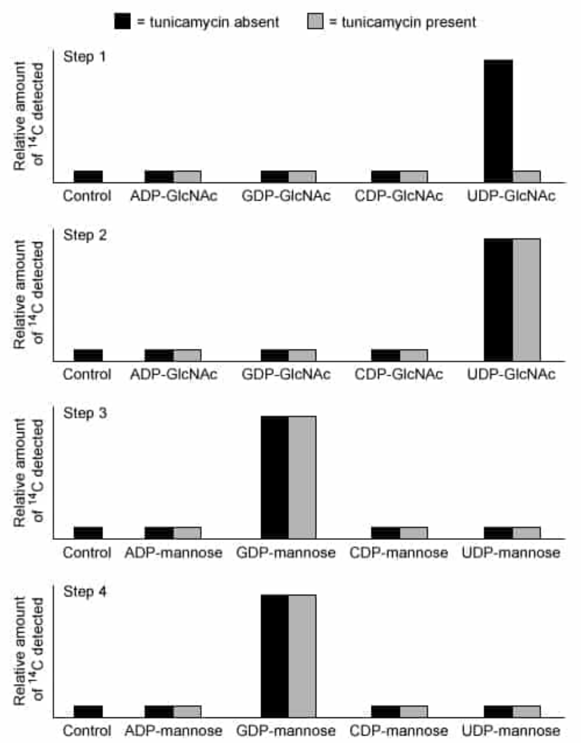


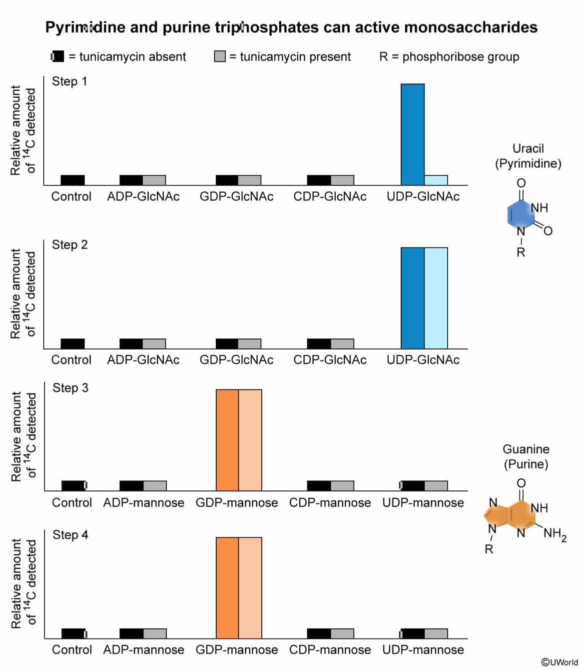
Figure 3 ^{14}C levels detected in liposomes incubated with various activated monosaccharides, with or without tunicamycin

The NTPs that activate GlcNAc and mannose can be classified as which of the following?

- A. Purine triphosphates [12%]
- B. Pyrimidine triphosphates [29%]
- C. A purine triphosphate and a pyrimidine triphosphate, respectively [11%]
- ✔ D. A pyrimidine triphosphate and a purine triphosphate, respectively [47%]

Correct
47% answered correctly

Explanation:



Nucleoside triphosphates (NTPs) can be classified based on the structure of the **nitrogenous base**. For NTPs, the four possible bases are adenine, guanine, cytosine, and uracil (in dNTPs, uracil is replaced by thymine). **Adenine and guanine** are both similar in structure to the molecule purine and are therefore classified as purine nucleotides, whereas **cytosine and uracil** are similar in structure to pyrimidine and are classified as pyrimidine nucleotides. NTPs are often used to activate other molecules for use in anabolic reactions.

Figure 3 shows the results of incubating various NDP-carbohydrates with liposomes (ie, small synthetic vesicles composed of phospholipids and other lipids). The liposomes contain dolichol phosphate linked to a particular precursor carbohydrate, to which the next monosaccharide in the glycan chain can be linked. However, the enzyme that catalyzes the linkage **only works with monosaccharides that are bonded to a specific NDP**, and this bond is produced when the **monosaccharide reacts with the corresponding NTP**.

In the experiment described by Figure 3, the monosaccharide in each NDP-monosaccharide molecule contains a ^{14}C atom at the anomeric carbon. This radioactive heavy isotope of carbon can be detected by autoradiography. If liposomes are found to be linked to molecules containing ^{14}C at significantly higher levels than control, this suggests that the monosaccharide is linked to the chain and therefore that the NDP-monosaccharide in the reaction could be used by the enzyme. The graphs show that ^{14}C was detected at levels above control (ie, no NDP-carbohydrate added) only when **UDP-GlcNAc or GDP-mannose** were used. Therefore, the NTPs that activate GlcNAc and mannose are **UTP (a pyrimidine triphosphate)** and **GTP (a purine triphosphate)**, respectively.

(Choices A and B) Only the nucleotide used to activate GlcNAc (UTP) is a pyrimidine triphosphate, and only the nucleotide used to activate mannose (GTP) is a purine triphosphate.

(Choice C) This answer would be correct if GlcNAc used GTP and mannose used UTP for activation, but the reverse is true.

Educational objective:
Nucleoside triphosphates (NTPs) can be classified based on the identities of their nitrogenous bases as either purines (adenine or guanine) or pyrimidines (cytosine or uracil). NTPs are commonly used to activate other molecules for anabolic reactions.

Time spent:0 sec	**QID**:404071
Subject: Biochemistry	**System:** Carbohydrates, Nucleotides, and Lipids

Most secretory and membrane-bound proteins participate in the *N*-linked glycosylation pathway, in which a complex polysaccharide group (also called a glycan) is linked to one or more asparagine residues on proteins in the lumen of the endoplasmic reticulum (ER). Over 30 congenital disorders are associated with defects in the enzymes responsible for the synthesis of *N*-linked glycans, and each disorder has a broad range of symptoms including eye, neuron, and bone pathologies.

The glycan is assembled on a dolichol phosphate molecule embedded in the ER membrane and consists primarily of *N*-acetylglucosamine (GlcNAc) and mannose residues. The anomeric carbon of each monosaccharide forms a glycosidic bond with the growing chain, and the full glycan is then transferred to a protein as a single unit. The first four steps of glycan synthesis are depicted in Figure 1.

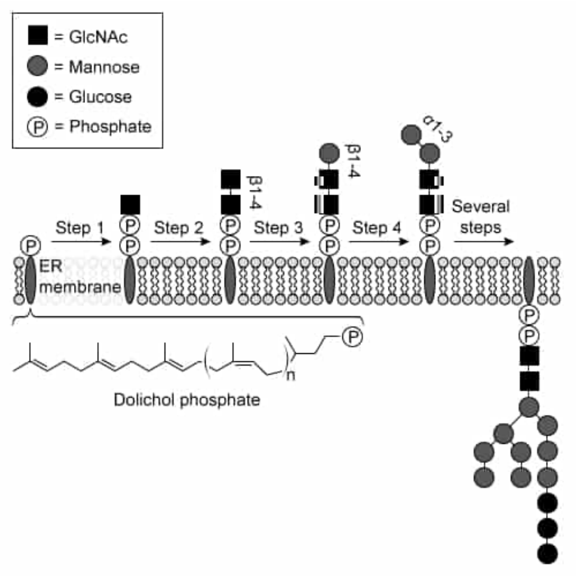


Figure 1 The first four steps of glycan synthesis in the *N*-linked glycosylation pathway and the full glycan. The structure of dolichol phosphate and linkages between carbohydrates are also shown.

Before a monosaccharide can be incorporated into the glycan, it must be activated by reacting with a nucleotide triphosphate (NTP) to form an NDP-carbohydrate molecule. The general mechanisms for formation of NDP-GlcNAc and NDP-mannose are shown in Figure 2, using the six-carbon sugar fructose 6-phosphate as the starting material.

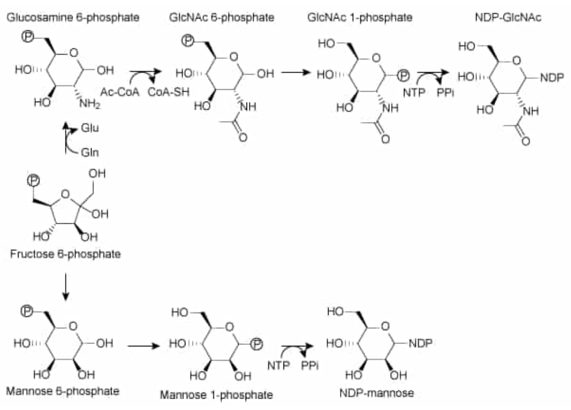


Figure 2 Formation of NDP-GlcNAc and NDP-mannose from fructose 6-phosphate

To determine which NDP activates each monosaccharide, researchers tested each step individually by incubating liposomes linked to precursor glycans, the appropriate transferase enzyme, and different NDP-carbohydrate molecules. Each test was done in the presence and absence of the drug tunicamycin. In each case, the anomeric carbon of the activated monosaccharide was a ^{14}C atom. After incubation, the liposomes were purified and analyzed by TLC and autoradiography. The results for the first four steps of the pathway are shown in Figure 3.

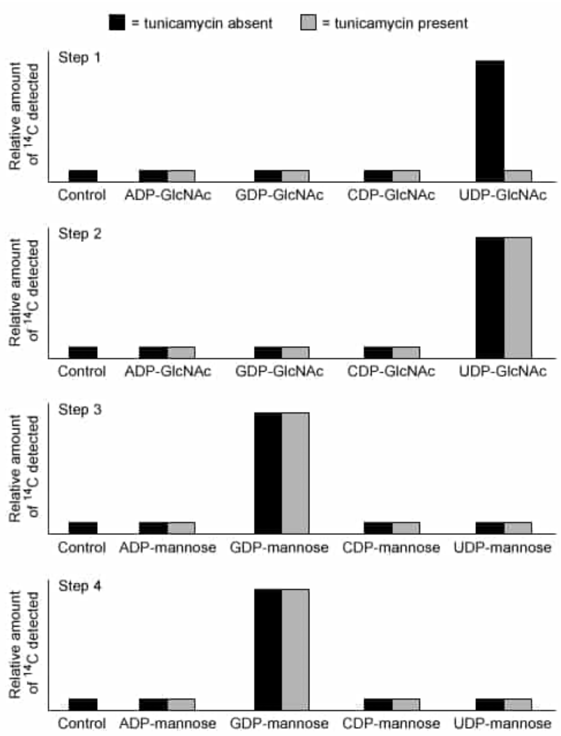


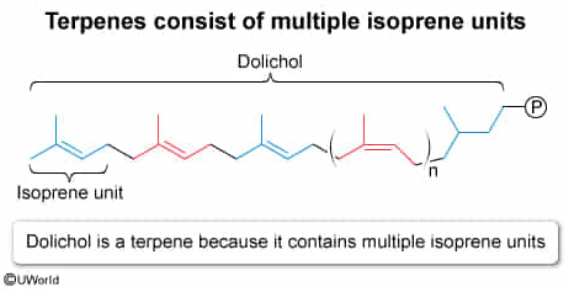
Figure 3 ^{14}C levels detected in liposomes incubated with various activated monosaccharides, with or without tunicamycin

Based on Figure 1, the dolichol portion of dolichol phosphate is an example of:

- A. an isoprene unit. [33%]
- B. a terpene. [51%]
- C. a sterol. [5%]
- D. a fatty acid. [9%]

Correct
52% answered correctly

Explanation:



Complex biological molecules are commonly assembled from simpler, smaller molecules. **Isoprenes are branched five-carbon molecules** that are the basis for cholesterol synthesis, and they can also be used in synthesis of other related molecules. When **two or more isoprene units** are linked, they **form a terpene**. For example, two isoprenes form a monoterpene, and four isoprenes form a diterpene.

The dolichol molecule shown in Figure 1 consists of **several isoprene units linked together**, largely in a repeating pattern. Because multiple isoprenes were linked together to form dolichol, **dolichol is an example of a terpene**.

(Choice A) As shown in Figure 1, dolichol contains *multiple* isoprene units and more than five carbons, so it is *not* a single isoprene unit.

(Choice C) Sterols such as cholesterol are formed from terpenes, but dolichol is not a sterol because it does not have the characteristic configuration of three six-membered rings and one five-membered ring.

(Choice D) A fatty acid is a hydrocarbon chain linked to a carboxylic acid group. Dolichol does not contain a carboxylic acid group.

Educational objective:
Isoprenes are branched five-carbon units. When two or more isoprene units are linked together, they form a terpene.

Time spent:0 sec
Subject:
Biochemistry

QID:404072
System:
Carbohydrates, Nucleotides, and Lipids

Most secretory and membrane-bound proteins participate in the *N*-linked glycosylation pathway, in which a complex polysaccharide group (also called a glycan) is linked to one or more asparagine residues on proteins in the lumen of the endoplasmic reticulum (ER). Over 30 congenital disorders are associated with defects in the enzymes responsible for the synthesis of *N*-linked glycans, and each disorder has a broad range of symptoms including eye, neuron, and bone pathologies.

The glycan is assembled on a dolichol phosphate molecule embedded in the ER membrane and consists primarily of *N*-acetylglucosamine (GlcNAc) and mannose residues. The anomeric carbon of each monosaccharide forms a glycosidic bond with the growing chain, and the full glycan is then transferred to a protein as a single unit. The first four steps of glycan synthesis are depicted in Figure 1.

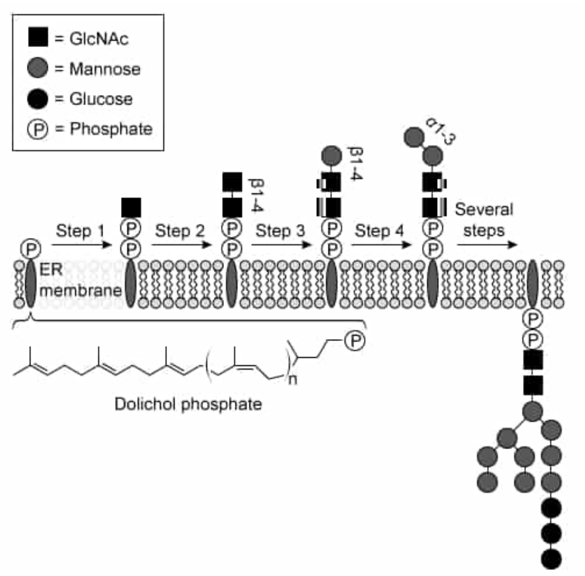


Figure 1 The first four steps of glycan synthesis in the *N*-linked glycosylation pathway and the full glycan. The structure of dolichol phosphate and linkages between carbohydrates are also shown.

Before a monosaccharide can be incorporated into the glycan, it must be activated by reacting with a nucleotide triphosphate (NTP) to form an NDP-carbohydrate molecule. The general mechanisms for formation of NDP-GlcNAc and NDP-mannose are shown in Figure 2, using the six-carbon sugar fructose 6-phosphate as the starting material.

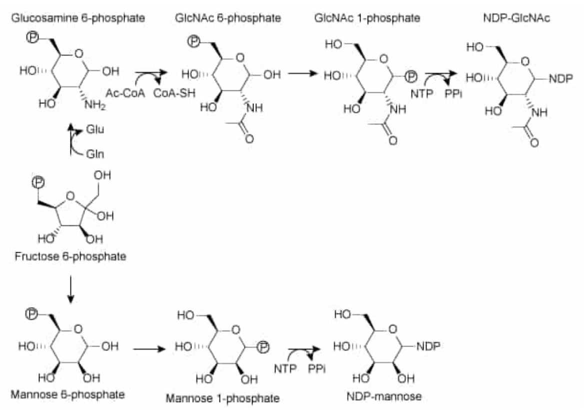


Figure 2 Formation of NDP-GlcNAc and NDP-mannose from fructose 6-phosphate

To determine which NDP activates each monosaccharide, researchers tested each step individually by incubating liposomes linked to precursor glycans, the appropriate transferase enzyme, and different NDP-carbohydrate molecules. Each test was done in the presence and absence of the drug tunicamycin. In each case, the anomeric carbon of the activated monosaccharide was a ^{14}C atom. After incubation, the liposomes were purified and analyzed by TLC and autoradiography. The results for the first four steps of the pathway are shown in Figure 3.

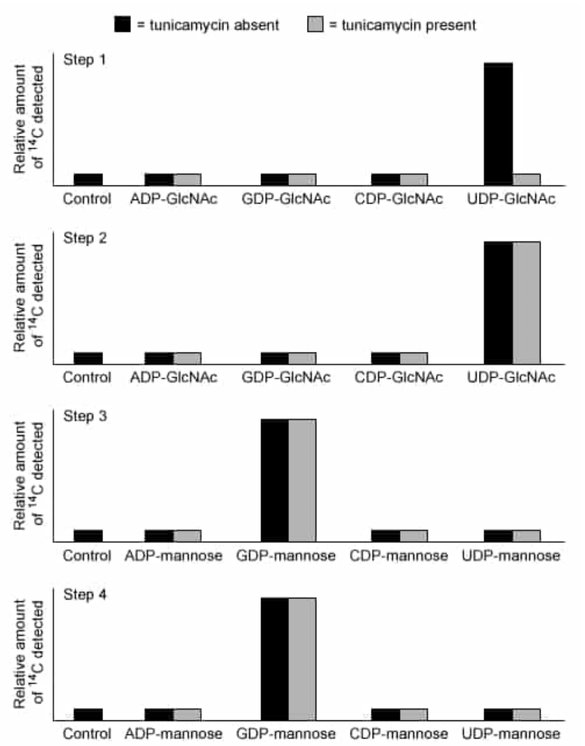


Figure 3 ^{14}C levels detected in liposomes incubated with various activated monosaccharides, with or without tunicamycin

Based on the information in Figures 1 and 3, tunicamycin inhibits which step of glycan synthesis?

- ✔ A. Step 1 [78%]
- B. Step 2 [7%]
- C. Step 3 [8%]
- D. Step 4 [5%]

Correct
79% answered correctly

Explanation:

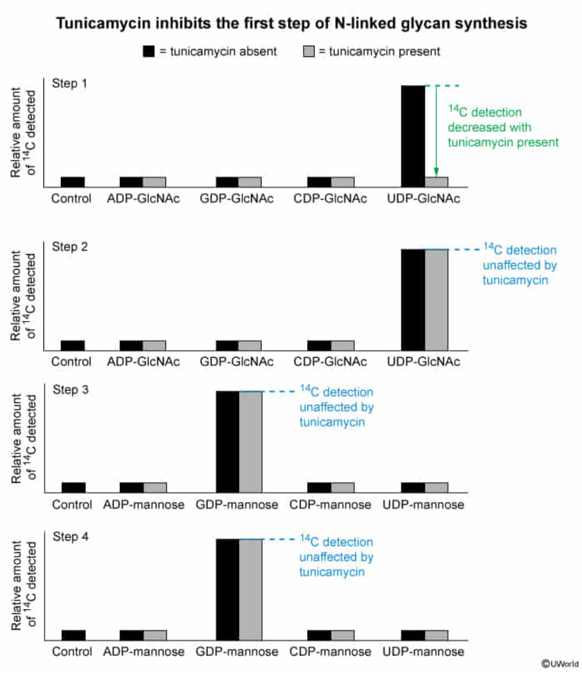


Figure 1 of the passage shows the first four steps of glycan synthesis. In each step, a single monosaccharide is added to the growing glycan chain: GlcNAc in the first two steps and mannose in the next two steps. Figure 3 shows incorporation of ^{14}C-containing monosaccharides into the growing glycan under different conditions. In each test, the necessary precursor glycan (if any) was linked to a liposome, and the next monosaccharide (linked to various NDPs) was added to the solution.

The graph for Step 1 shows that GlcNAc is successfully linked to the liposome when UDP-GlcNAc is used as the substrate, but that when tunicamycin is added, GlcNAc is not successfully transferred regardless of what substrate is used. Therefore, **tunicamycin must inhibit step 1 of the pathway**.

(Choices B, C, and D) Each of these graphs shows that the appropriate sugar is successfully added to the growing glycan when UDP-GlcNAc (Step 2) or GDP-mannose (Steps 3 and 4) are used as substrates. This remains true whether tunicamycin is present or not, so tunicamycin does not inhibit any of these steps.

Educational objective:
Inhibition of a step in a biological pathway can be tested by examining each step of the pathway in the presence and absence of the drug.

Time spent:0 sec
Subject:
Biochemistry

QID:404073
System:
Carbohydrates, Nucleotides, and Lipids

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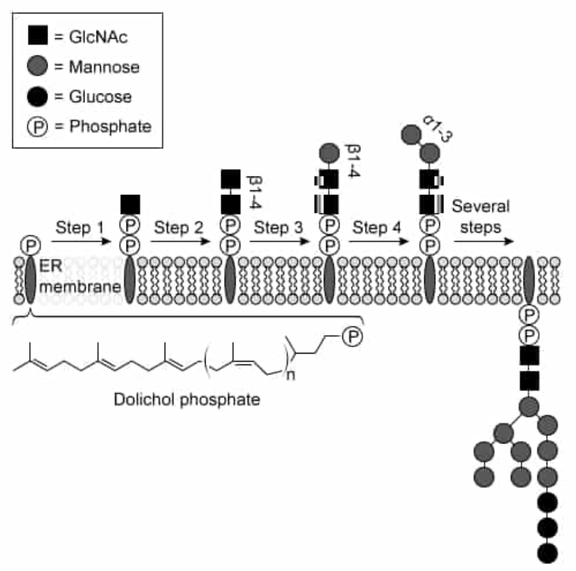

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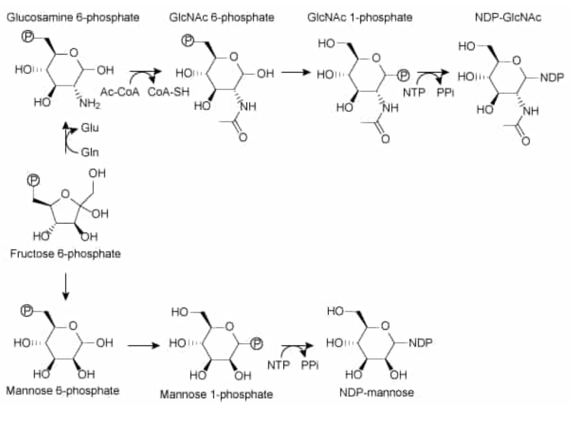

Figure 2 Formation of NDP-GlcNAc and NDP-mannose from fructose 6-phosphate

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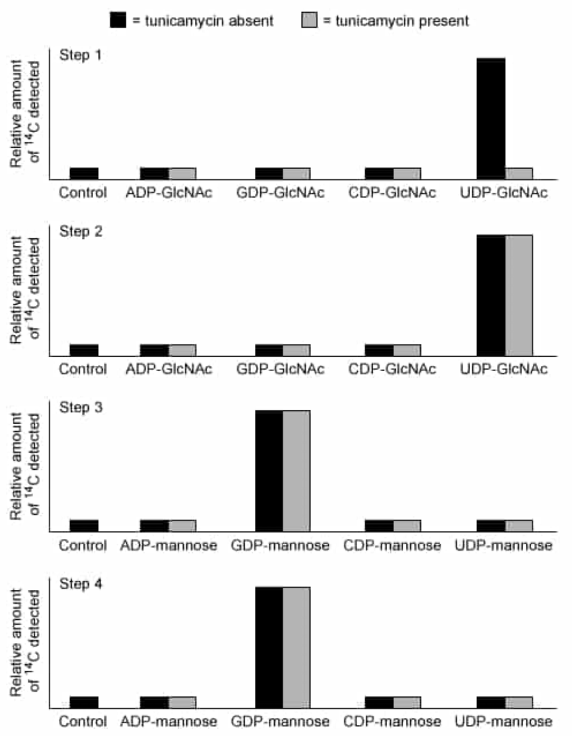


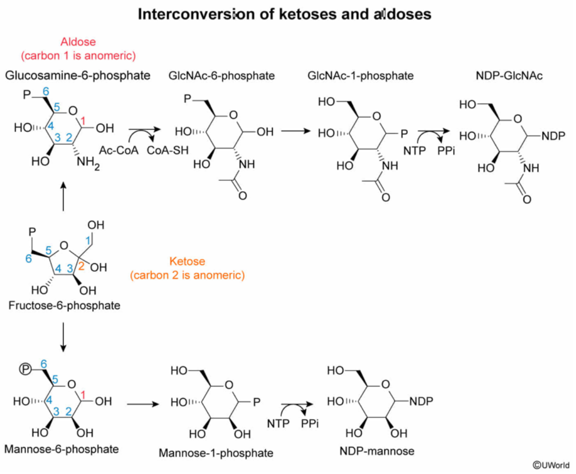
Figure 3 ^{14}C levels detected in liposomes incubated with various activated monosaccharides, with or without tunicamycin

Based on Figure 2, the conversion of fructose 6-phosphate to glucosamine 6-phosphate involves which of the following steps?

- A. Addition of a carbon atom converts a pentose derivative to a hexose derivative [31%]
- B. Loss of a carbon atom converts a hexose derivative to a pentose derivative [1%]
- C. Movement of the anomeric carbon from position 1 to position 2 converts an aldose derivative to a ketose derivative [15%]
- ✔ D. Movement of the anomeric carbon from position 2 to position 1 converts a ketose derivative to an aldose derivative [50%]

Correct
50% answered correctly

Explanation:



Monosaccharides can be classified by the number of carbons they contain. In this classification system, a carbohydrate is assigned a Greek prefix indicating the number of carbons it contains (eg, *tri*–, *tetr*–, *pent*–) followed by the suffix *–ose*. For example, a **five-carbon sugar is a pentose** and a **six-carbon sugar is a hexose**.

In addition, monosaccharides are also classified by the position of the anomeric carbon. When the anomeric carbon is on carbon 1, the linear form of the sugar contains an aldehyde group. Consequently, sugars with the anomeric carbon at carbon 1 are called aldoses. Although the aldehyde group becomes a hemiacetal when the sugar cyclizes, the sugar is still called an aldose in cyclic form. In contrast, when the anomeric carbon is on carbon 2, the linear form of the sugar contains a ketone (or a hemiketal in the cyclic form), and the sugar is called a ketose.

Figure 2 shows that fructose 6-phosphate and glucosamine 6-phosphate each have six carbons, making both of them hexose derivatives. However, the **anomeric carbon of fructose 6-phosphate is on carbon 2**, whereas the anomeric carbon of glucosamine 6-phosphate is on **carbon 1**. Therefore, during conversion of fructose 6-phosphate to glucosamine 6-phosphate, a **ketose derivative** (fructose 6-phosphate) is **converted to an aldose derivative** (glucosamine 6-phosphate).

(Choices A and B) As stated in the passage, fructose 6-phosphate is a six-carbon sugar (ie, a hexose). Figure 2 shows that no carbons are gained or lost as fructose 6-phosphate becomes glucosamine 6-phosphate, so no interconversion between pentoses and hexoses occurs in this pathway. Note that the change from a five-membered *ring* in fructose 6-phosphate to a six-membered *ring* in glucosamine 6-phosphate is a change from a furanose to a pyranose, *not* a pentose to a hexose.

(Choice C) Fructose 6-phosphate is a ketose derivative, not an aldose derivative, and glucosamine 6-phosphate is an aldose derivative, not a ketose derivative.

Educational objective:
Monosaccharides are classified by the number of carbons they contain (eg, triose, tetrose, pentose, hexose) and the position of the anomeric carbon (ie, ketose, aldose).

Time spent:0 sec

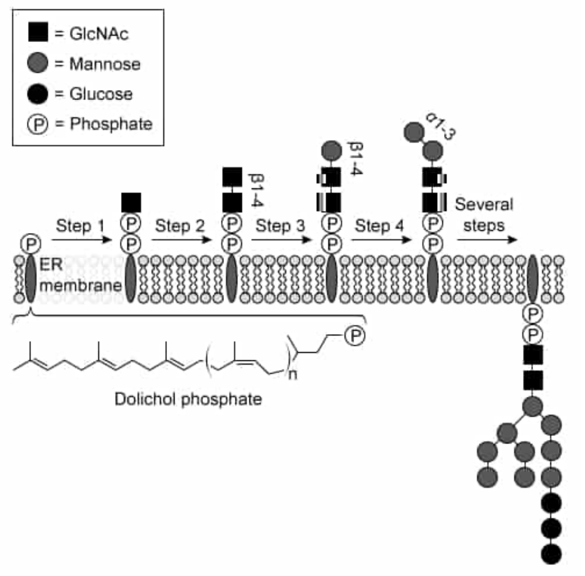
Subject:

QID:404074

System:
Carbohydrates, Nucleotides, and Lipids

Most secretory and membrane-bound proteins participate in the *N*-linked glycosylation pathway, in which a complex polysaccharide group (also called a glycan) is linked to one or more asparagine residues on proteins in the lumen of the endoplasmic reticulum (ER). Over 30 congenital disorders are associated with defects in the enzymes responsible for the synthesis of *N*-linked glycans, and each disorder has a broad range of symptoms including eye, neuron, and bone pathologies.

The glycan is assembled on a dolichol phosphate molecule embedded in the ER membrane and consists primarily of *N*-acetylglucosamine (GlcNAc) and mannose residues. The anomeric carbon of each monosaccharide forms a glycosidic bond with the growing chain, and the full glycan is then transferred to a protein as a single unit. The first four steps of glycan synthesis are depicted in Figure 1.



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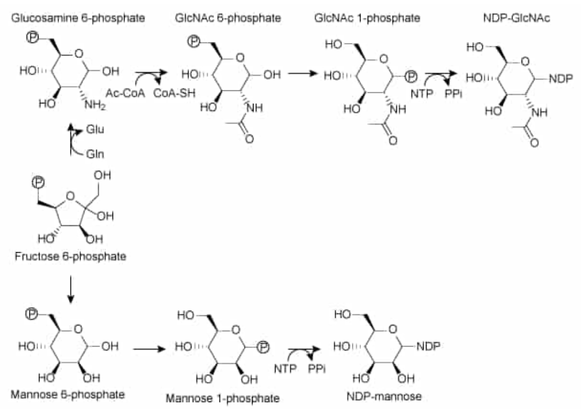

Figure 2 Formation of NDP-GlcNAc and NDP-mannose from fructose 6-phosphate

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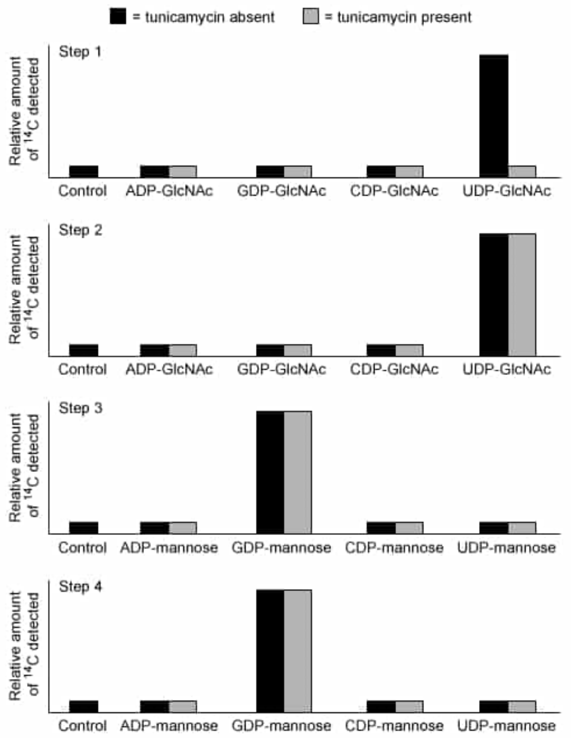


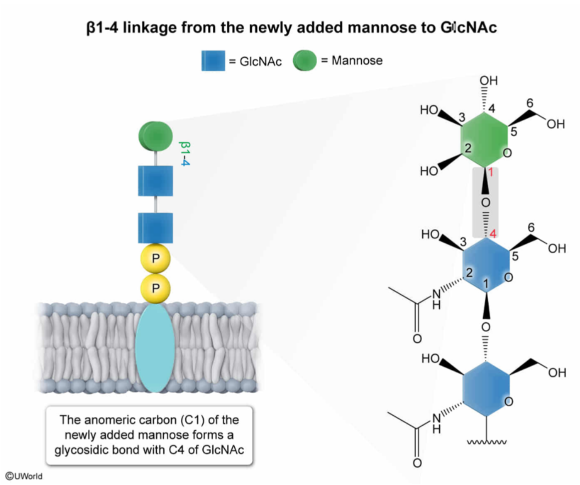
Figure 3 ^{14}C levels detected in liposomes incubated with various activated monosaccharides, with or without tunicamycin

Based on Figure 1, a GlcNAc residue in the glycan chain is linked to a mannose residue by:

- ○ A. a glycosidic bond between carbon 1 of GlcNAc and carbon 4 of mannose. [34%]
- ✔ ◉ B. a glycosidic bond between carbon 1 of mannose and carbon 4 of GlcNAc. [56%]
- ○ C. a glycosidic bond between carbon 1 of GlcNAc and carbon 3 of mannose. [5%]
- ○ D. a glycosidic bond between carbon 1 of mannose and carbon 3 of GlcNAc. [3%]

Correct
55% answered correctly

Explanation:



The individual monosaccharides in a polysaccharide are linked to each other by glycosidic bonds. In a polysaccharide, each glycosidic bond links the anomeric carbon of one sugar to any carbon of the other sugar. Glycosidic bonds can also form between the anomeric carbon of a sugar and any other biomolecule (eg, amino acids, nucleotides).

The connection between sugars in a glycosidic bond is described by listing the configuration of the anomeric carbon (α or β), the number of the anomeric carbon (usually carbon 1), and the number of the linked carbon in the other sugar.

Figure 1 shows that a GlcNAc residue and a mannose residue in the glycan chain are connected by a β1-4 linkage, meaning that carbon 1 of one of the monosaccharides is in the β configuration and is linked to carbon 4 of the other monosaccharide. According to the passage, each carbohydrate is added to the growing chain when the **anomeric carbon of the incoming monosaccharide** forms a glycosidic bond with the chain. Because **mannose is added *after*** GlcNAc, the anomeric carbon of mannose (the incoming monosaccharide) must be involved in the glycosidic bond. Therefore, GlcNAc and mannose are linked by a glycosidic bond between **carbon 1 of mannose** and **carbon 4 of GlcNac**.

(Choice A) Carbon 1 of the GlcNAc residue of interest is already involved in a glycosidic bond with the other GlcNAc residue in the chain and cannot be involved in the linkage to mannose.

(Choices C and D) The two mannose residues in the chain are linked to each other through a glycosidic bond between carbon 1 of one mannose and carbon 3 of the other, but the question asks for the linkage between *GlcNAc* and mannose.

Educational objective:
Glycosidic linkages between monosaccharide units can be described by listing the configuration of the anomeric carbon involved in the bond, its number, and the number of the carbon in the other monosaccharide (eg, α1-3 linkage, β1-4 linkage).

Time spent:0 sec
Subject:
Biochemistry

QID:404075
System:
Carbohydrates, Nucleotides, and Lipids

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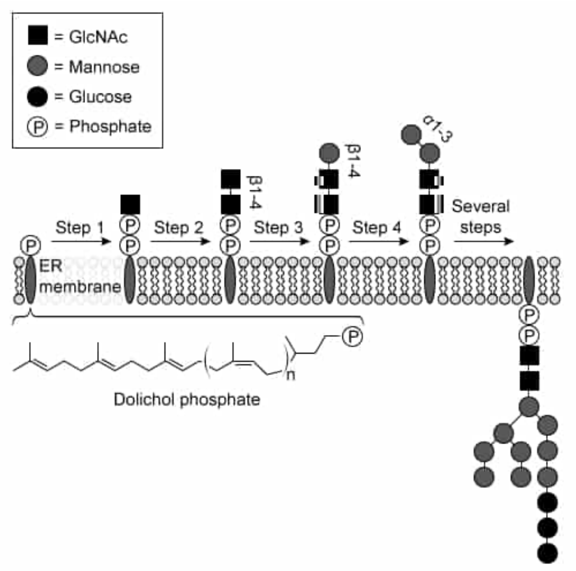

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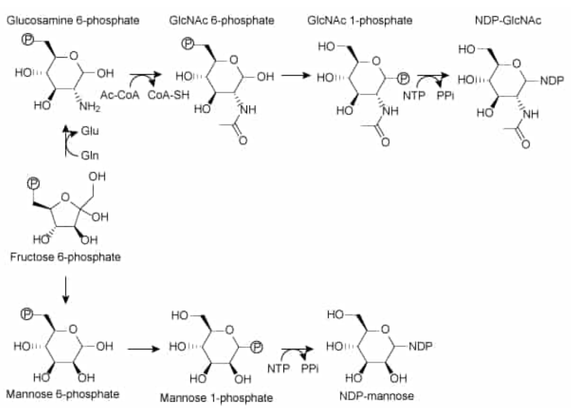


Figure 2 Formation of NDP-GlcNAc and NDP-mannose from fructose 6-phosphate

To determine which NDP activates each monosaccharide, researchers tested each step individually by incubating liposomes linked to precursor glycans, the appropriate transferase enzyme, and different NDP-carbohydrate molecules. Each test was done in the presence and absence of the drug tunicamycin. In each case, the anomeric carbon of the activated monosaccharide was a ^{14}C atom. After incubation, the liposomes were purified and analyzed by TLC and autoradiography. The results for the first four steps of the pathway are shown in Figure 3.

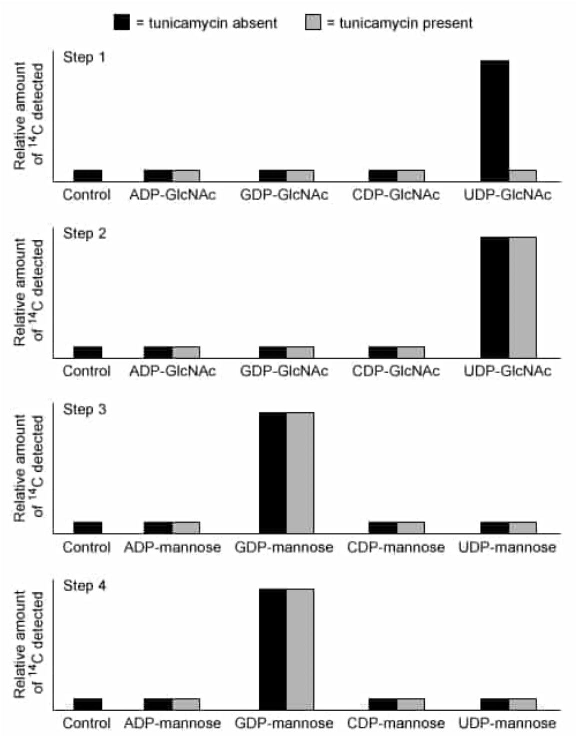


Figure 3 ^{14}C levels detected in liposomes incubated with various activated monosaccharides, with or without tunicamycin

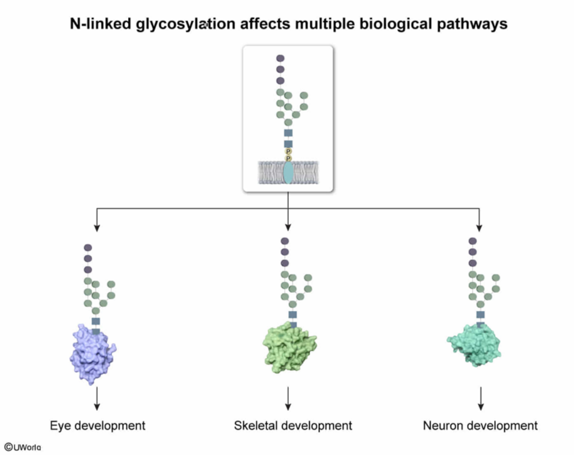
Which of the following best explains why disorders affecting the enzymes of *N*-linked glycan synthesis result in a broad range of symptoms?

- A. NDP-mannose and NDP-GlcNAc are both synthesized from fructose 6-phosphate, which cannot be synthesized when the enzymes of *N*-linked glycan synthesis are defective. [11%]
- B. Each disorder prevents monosaccharides from becoming activated, so they cannot be transferred to the glycan. [5%]
- C. Transfer of an incomplete glycan to a protein causes the protein to fold incorrectly, leading to overactive lysosomal protein degradation in multiple organ systems. [24%]
- D. The same glycan is transferred to many different proteins, so failure of any step of *N*-linked glycan synthesis affects multiple pathways. [59%]

Correct

59% answered correctly

Explanation:



Different biochemical pathways are required in different cell types, tissues, and organs to varying degrees. A given pathway may also have increased or decreased importance at different times in an organism's life cycle. Defects in one or more enzymes of a particular pathway therefore primarily affect the organs and tissues that require that pathway and only at the times when those pathways are required. For example, if an enzyme in a pathway required for eye development is defective, the eye will likely not develop correctly.

According to the passage, most secretory and membrane-bound proteins participate in *N*-linked glycosylation. All eukaryotic cells contain membrane-bound proteins in both their plasma and organelle membranes, and many cell types secrete proteins. The passage also indicates that many of these proteins are modified by the **same glycan** (the *N*-linked glycan), which is transferred from the ER membrane to each protein as a single unit.

If any enzyme involved in synthesis of the *N*-linked glycan is defective, the glycan will likely either not be fully assembled or transferred at all, or assembly and transfer will happen too slowly. Either case will affect *all* proteins that undergo *N*-linked glycosylation. **Because these proteins are present in many cell types** (and therefore many tissues and organs), all **corresponding tissues are likely to be affected** to some degree, resulting in a **broad range of pathologies**.

(Choice A) Fructose 6-phosphate is synthesized by many pathways, including glycolysis, gluconeogenesis, and the pentose phosphate pathway.

(Choice B) Activation of monosaccharides represents only *some* of the steps in the *N*-linked glycosylation pathway. Many other steps are also required in addition to those shown in Figure 1.

(Choice C) Although protein glycosylation is involved in correct protein folding and incorrectly folded proteins are commonly sent to the lysosome, the passage states that the *fully assembled* glycan is transferred from the ER membrane to proteins as a *single unit*. Therefore, incomplete glycans most likely remain on the ER membrane and are *not* transferred to proteins.

Educational objective:
Defects in biological pathways are likely to affect all cell types, tissues, and organs in which those pathways are involved.

Time spent: 0 sec	**QID**:404076
Subject: Biochemistry	**System:** Carbohydrates, Nucleotides, and Lipids